Discussion Agenda 03/06, 04/09, 05/09, 06/14, 07/19/2019 (CGM+Clumps)

Below is an ongoing discussion agenda / live progress report for AGORA Paper Groups "CGM" and "Clumps" for semi-monthly web conferences.

Contents

Most recent changes that require your attention are in red color!

(1st revision: 02/26/2019, 2nd revision 04/09/2019, 3rd revision 05/09/2019)

1. Paper Group Progress Report (as of June 2019)

(1) Science-oriented Projects

<table>
<tr><th></th><th>Paper "CGM"</th><th>Paper "Clumps"</th><th>Paper "Metallicity"</th></tr>
<tr><td>GOALS</td><td>• Run ~1e12 Ms halo to study absorption lines</td><td>• Run ~1e12 (~1e13) Ms halo to study SF clumps at high z</td><td>• Run ~1e12 Ms halo to study metal distribution</td></tr>
<tr><td>LEADERS</td><td>• Cameron Hummels (Caltech)
• Santi Roca-Fabrega (UCM)</td><td>• Daniel Ceverino (Heidelberg)
• Alessandro Lupi (IAP)</td><td></td></tr>
<tr><td>PARTICIPANTS</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Dekel, Mandelker, Primack, Roca-Fabrega, Strawn</td></tr>
<tr><td>ART-II</td><td>(Feldmann)</td></tr>
<tr><td>CHANGA</td><td>(Quinn)</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim (Ji-hoon), Kim (Ki-won), O'Shea, Shin, Wise</td></tr>
<tr><td>GADGET</td><td>Nagamine, Shimizu</td></tr>
<tr><td>GASOLINE</td><td>(Madau, Shen, Wadsley, Wissing)</td></tr>
<tr><td>GEAR</td><td>Hausammann, Revaz</td></tr>
<tr><td>GIZMO</td><td>Dong, Hopkins, Hummels</td></tr>
<tr><td>RAMSES</td><td>Agertz, Roca-Fabrega, Teyssier</td></tr>
</table>
(Code leaders as of July 2019)</td>
<td>• Code representatives:
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<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Dekel, Lee, Mandelker, Primack, Roca-Fabrega</td></tr>
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<tr><td>RAMSES</td><td>Agertz, Roca-Fabrega</td></tr>
</table>
(Code leaders as of July 2019)
- ICs help: Buehlmann, Hahn
- Analysis help: Chakrabarti, Kim, Turk, Roca-Fàbrega, Strawn</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>GADGET</td><td>Chakrabarti, Lewis</td></tr>
<tr><td>GEAR</td><td>(Revaz)</td></tr>
<tr><td>RAMSES</td><td>(Pallottini, Teyssier)</td></tr>
</table>
- Analysis help: Kim

• Working Groups engaged:
- IV (common analysis)
- IX (char. of cos. galaxies)
- XII (interstellar medium)</td></tr>
</table>

	– ICs help: Buehlmann, Hahn, Simpson – Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Madau, Roca-Fabrega, Smith, Turk, Strawn • Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – V (subgrid physics) – IX (char. of cos. galaxies) – X (outflows)	• Working Groups engaged: – III (common cosmo ICs) – IV (common analysis) – XI (high-z galaxies)	
PROGRESS MADE OR BEING MADE	• Visit **Paper's Workspace** for the latest updates including issues being faced by the group and the individual code groups, and the Overleaf draft being written. • Code leaders have submitted preliminary datasets for the proposed cosmological run aimed at calibrating the feedback at z=4 first. Composite figures in the **Paper "CGM" Workspace**. • Strawn has provided an **example script** using yt+Trident to produce the absorption lines, HI covering fraction, heat maps on radius - column density plane, etc. • Hummels has updated the **ipython notebook** that explains how to use yt+Trident on datasets from various codes. Trident is now fully functional with "demeshened" version of yt (i.e., **yt-4.0** branch) as of Jun. 2018. More information in the **Paper "CGM" Workspace**.	• Visit **Paper's Workspace** for the latest updates, and the Overleaf draft being written. • **Task Force** participants have completed or are about to complete the action items to jointly launch Papers "CGM" and "Clumps".	• Visit **Paper's Workspace** for the latest updates. • Will piggyback on the planned Paper "CGM" runs and evolve them further down to z=0.
OTHER LINKS	• **Paper Workspace** • **Trident download and documentation** • **Trident method paper (Hummels et al. 2016)**	• **Paper Workspace**	• **Paper Workspace**

Cf. other science-oriented project ideas: see **this link**

(2) Previous Projects

<table>
<tr><th></th><th>Paper "DM-Highres"</th><th>Paper "NoSubgrid"</th><th>Paper "Disk-II"</th></tr>
<tr><td>GOALS</td><td>• Run Proof-of-concept-DM (~1e11 Ms halo) at higher res</td><td>• Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)</td><td>• Compare simulation codes using common disk ICs</td></tr>
<tr><td>LEADERS</td><td>• Oliver Hahn (OCA, Nice)
• Christine Simpson (HITS)</td><td>• Nick Gnedin (Fermilab)
• Piero Madau (UCSC)
• Britton Smith (SDSC)</td><td>• Santi Roca-Fabrega (UCM)</td></tr>
<tr><td>PARTICIPANTS</td>
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<tr><td>ART-I</td><td>Ceverino, Klypin, Primack</td></tr>
<tr><td>ART-II</td><td>Gnedin, Feldmann, Li, Semenov (Leitner)</td></tr>
<tr><td>CHANGA</td><td>Quinn, Wallace</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim, Simpson, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine, Onorbe, (Rocha)</td></tr>
<tr><td>GADGET-AFS/CFS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Madau, Shen, (Pillepich, Zolotov)</td></tr>
<tr><td>GCD+</td><td>Kawata</td></tr>
<tr><td>GIZMO-AFS</td><td>Garrison-Kimmel, Lupi, Hopkins, Onorbe</td></tr>
<tr><td>PKDGRAV2</td><td>(Guedes, Kuhlen)</td></tr>
<tr><td>RAMSES</td><td>Hahn, Roca-Fabrega, Teyssier</td></tr>
</table>
(Code leaders as of Aug. 2016)
- Others interested: Dekel, Feldmann, Kawata, Pallottini, Robles</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Klypin</td></tr>
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<tr><td>GADGET</td><td>Choi, Nagamine</td></tr>
<tr><td>GADGET-SPHS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Keller, Madau, Mayer, Shen, Wadsley</td></tr>
<tr><td>GEAR</td><td>Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Thompson</td></tr>
<tr><td>RAMSES</td><td>Agertz, Butler, Teyssier</td></tr>
</table>
(Grackle implementers acknowledged in Paper "4")</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Arraki, Ceverino, Klypin, Mandelker, Snyder</td></tr>
<tr><td>ART-II</td><td>Feldmann, Gnedin, Leitner, Li, Semenov</td></tr>
<tr><td>CHANGA</td><td>Quinn, Wallace</td></tr>
<tr><td>ENZO</td><td>Abel, Butler, Butsky, Forbes, Hummels, Kim, O'Shea, Simpson, Turk, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine, Chakrabarti, Shimizu, (Todoroki)</td></tr>
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<tr><td>RAMSES</td><td>Agertz, Butler, DeGraf, Teyssier, Nickerson</td></tr>
</table>
(Code leaders as of Aug. 2016 for Paper "4" before)</td></tr>
<tr><td>PROGRESS MADE OR BEING MADE</td>
<td>• Visit Paper's Workspace where Simpson, Hahn and others have been updating the progress.

• We will plan to have 5-code comparison. As of Aug. 2017, 4+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC.

• We may focus on the (survival of) substructures. This subset of topics could potentially to be led by Jiang</td>
<td>• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.</td>
<td>• Visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper - previously known as Paper "4"). The agora-analysis-script repository has been accordingly updated.

• Analysis items that were not covered in our first disk comparison paper (see this link) will be considered in the next paper, Paper "Disk-II".</td></tr>
</table>

	as a smaller paper. • Consistent tree – python interface has been built by Britton Smith (ytree).		• A subset of topics could be discussed in the "Metal Diffusion" or "Secular evolution" paper (see this link).
OTHER LINKS	• Progress report in Paper's Workspace and/or in WG III's Workspace. • MUSIC Google group • ytree documentation	• GRACKLE Google group • GRACKLE download & documentation • GRACKLE method paper (Smith et al. 2017) • AGORA API for GRACKLE by Nick Gnedin	• Progress report in Paper's Workspace and/or in WG V's Workspace. • yt-AGORA Google group

2. News and Updates

(1) AGORA Workshop 2019

• As always, the concluding day (Friday) of the 2019 Santa Cruz Galaxy Workshop (Aug. 5-9, 2019) will be the first day of the 8th AGORA Workshop, Aug. 9-10, 2019. We thank the continuous logistical support by Joel Primack and UC Santa Cruz. More information in your mailbox (agora mailing list) or in this **blog post**.

3. Recaps

(1) AGORA SMBH Prescription for Accretion and Feedback:

• Please visit **this link**.

(2) Strategy to Store Simulation Outputs:

• Please visit **this link**.

(3) How to Request an Account on NERSC:

• If you represent a code group but don't have a NERSC user account, please visit **this page**. Select "standard" account and find the repository name "agora". Joel Primack will add your name to the group when approved.
• We thank the annual effort by Joel Primack and Tom Abel to secure a continuous access to the NERSC computing resources for AGORA. The latest allocation for AY19 (Jan. 8, 2019 - Jan. 13, 2020) is 15 TB per project/archive storage each with half a million computing hours.

(4) Your Own yt Installation on NERSC:

• Please visit **this link**.

(5) yt-Sunrise Pipeline for AGORA on NERSC / Working Installation of Sunrise as Docker Image:

- Please visit **this link**.

(6) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the **yt-AGORA mini-workshop** at UCSC in Mar. 2014.

(7) Other Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: **here**
- Internal policy regarding the disclosure of AGORA simulation results: **here**
- Collection of softwares relevant to AGORA effort: **here**

4. Your Feedback

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the **mailing list**. You can also directly contact the **project coordinator** anytime!